# PROJECT REPORT

# TASK 16: LOAD A CSV DATASET USING PANDAS

## 1. Abstract

This project demonstrates how to load a Comma-Separated Values (CSV) dataset using Python's Pandas library. The program displays the first five and last five records and examines the dataset's dimensions, column names, data types, and missing values.

The project provides practical experience in importing structured data and inspecting it before performing further analysis.

## 2. Introduction

CSV files are widely used to store tabular data because they are simple to create, read, and process. They can contain information such as student records, employee details, sales transactions, and other structured data.

Pandas is a Python library designed for data manipulation and analysis. Its `read_csv()` function allows users to load CSV files into a DataFrame, which organizes information into rows and columns.

Inspecting a dataset after loading it is an important first step in data analysis because it helps users understand its structure and identify potential data-quality issues.

## 3. Objectives

The main objectives of this project are:

- To load a CSV dataset using Pandas.
- To display the first five records using the `head()` function.
- To display the last five records using the `tail()` function.
- To identify the number of rows and columns.
- To inspect column names and data types.
- To check for missing values.
- To understand how structured data can be prepared for analysis.

## 4. Tools and Technologies Used

### Python

Python is the programming language used to implement the CSV-loading workflow.

### Pandas

Pandas provides functions to read CSV files, inspect tabular data, and perform

basic data-quality checks.

### CSV

CSV is the file format used to store the sample dataset.

### Google Colab and Jupyter Notebook

Google Colab provides a browser-based Jupyter Notebook environment in which Python code can be executed interactively.

### GitHub

GitHub is used to store the source notebook and documentation and to share the completed project.

## 5. Dataset Description

The project uses a manually created sample dataset containing 12 student records.

The dataset contains five columns:

| Column Name | Description |
|---|---|
| student_id | Unique student identifier |
| name | Student name |
| department | Academic department |
| age | Student age |
| marks | Student marks |

The dataset is stored in a CSV file named `dataset.csv`.

The student records are fictional and are used only for demonstration purposes.

## 6. Methodology

The project follows these steps:

1. Create a sample CSV dataset.
2. Import the Pandas library.
3. Load the CSV file using `pd.read_csv()`.
4. Store the data in a Pandas DataFrame.
5. Display the first five records using `df.head()`.
6. Display the last five records using `df.tail()`.
7. Examine the number of rows and columns using `df.shape`.
8. Display column names using `df.columns`.
9. Inspect column data types using `df.dtypes`.
10. Count missing values using `df.isnull().sum()`.
11. Verify that the dataset has loaded correctly.

## 7. Pandas Functions Used

### 7.1 `pd.read_csv()`

This function reads a CSV file and converts its contents into a Pandas DataFrame.

### 7.2 `df.head()`

The `head()` function displays the first five records by default. It allows users to inspect the beginning of the dataset.

### 7.3 `df.tail()`

The `tail()` function displays the last five records by default. It allows users to inspect the end of the dataset.

### 7.4 `df.shape`

The `shape` property returns the number of rows and columns in the DataFrame.

### 7.5 `df.columns`

This property returns the names of the columns in the dataset.

### 7.6 `df.dtypes`

This property identifies the data type of each column, such as integers or text.

### 7.7 `df.isnull().sum()`

This expression counts missing values in each column. It helps determine whether the dataset requires further cleaning.

## 8. Results and Observations

After loading the supplied CSV file, the dataset contains 12 records and five columns.

The program displays the first five student records and the last five student records. It also displays the column names, data types, and missing-value counts.

The sample dataset contains no missing values.

The first five records begin with student ID 1001, while the last five records begin with student ID 1008 and end with student ID 1012.

These observations confirm that the dataset has been loaded and inspected successfully.

## 9. Program Output Evidence

**Screenshot 1: Program Output**

Insert a screenshot from Google Colab showing the output for:

- The number of rows and columns.
- The first five records.
- The last five records.
- Column names and data types.
- Missing-value counts.

**Screenshot 2: Dataset Preview**

Insert a screenshot showing `dataset.csv` or the loaded DataFrame.

## 10. Applications

CSV loading and dataset inspection are commonly used in:

- Student performance analysis.
- Sales and revenue analysis.
- Employee data analysis.
- Exploratory data analysis.
- Data cleaning and validation.
- Preparing structured datasets for machine-learning models.

## 11. Limitations

The project uses a small, manually created dataset. It demonstrates CSV loading and inspection but does not include advanced data cleaning, visualization, predictive modelling, or interactive file selection.

## 12. Future Enhancements

The project could be extended by:

- Loading larger datasets.
- Allowing users to upload their own CSV files.
- Identifying duplicate records.
- Handling missing values.
- Generating summary statistics and visualizations.
- Preparing datasets for machine-learning algorithms.

## 13. Conclusion

The project successfully demonstrates how Python and Pandas can be used to load and inspect a CSV dataset.

The use of `read_csv()`, `head()`, `tail()`, `shape`, `dtypes`, and `isnull().sum()` provides a basic understanding of structured data processing.

These skills are useful for more advanced data analysis and machine-learning tasks.

## 14. References

1. Pandas Documentation: https://pandas.pydata.org/docs/
2. Pandas `read_csv()` Reference:
https://pandas.pydata.org/docs/reference/api/pandas.read_csv.html
3. Python Documentation: https://docs.python.org/3/

## 15. Project Links

GitHub Repository: https://github.com/venky1318/Load_CSV_Dataset